

Binomial Distributions

Sampling distribution of a
Binomial

Example with two trials

Example with ten trials

L15: The Binomial Distribution

Today's objectives

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- ▶ Recognize when a count follows a Binomial distribution
- ▶ Interpret the parameters n and p
- ▶ Calculate exact binomial probabilities by hand and in R

Other types of outcomes

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We've now seen how we can use a normal probability distribution to help us evaluate continuous variables.

What about count variable based on repeated experiments where the outcome has only two possibilities?

The binomial setting and binomial distributions

- ▶ An elementary school administers eye exams to 800 students. How many students have perfect vision?
- ▶ A new treatment for pancreatic cancer is tried on 250 patients. How many survive for five years?
- ▶ You plant 10 dogwood trees. How many live through the winter?

What are the common threads to each of these questions?

- ▶ Something is done n number of times.
- ▶ The outcome of interest for each question is categorical (binary - two levels)

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Figure 1: Jacob Bernoulli

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- ▶ Bernoulli Random Variable: The variable must assume one of two possible mutually exclusive outcomes
- ▶ Each trial results in either a success or a failure
- ▶ Derived from the experiment: counting the number of occurrences of an event in n independent trials
- ▶ Random Variable: X = number of times the event happens in the fixed number of trials (n)
- ▶ Parameters
 - ▶ n = number of trials
 - ▶ p = probability of success (event happening)

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- ▶ The n trials are independent. Knowing the result of one trial does not change the probabilities of the two possible outcomes in other observations
- ▶ Each observation is either a “success” or a “failure” (usually coded 1 or 0). These terms are labels, not value judgments.
- ▶ The probability of success, denoted p , is the same for each trial.

$$P(0 \cup 1) = P(0) + P(1) = 1$$

Example 1

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A researcher has access to 40 men and 40 women and selects 10 of them at random, without replacement, to participate in an experiment. The number of women selected can be represented by X . Is X binomially distributed?

- ▶ Not exactly. Sampling without replacement makes the trials dependent and changes the probability of selecting a woman after each draw. The exact model is Hypergeometric.

Example 2

A pharmaceutical company inspects a simple random sample of 10 empty plastic containers from a shipment of 10,000. They are examined for traces of benzene. Suppose that 10% of the containers in the shipment contain benzene. Let X represent the number of containers contaminated with benzene. Is X binomially distributed?

- ▶ Issue: Each time you sample one bottle, it affects the chance that the next bottle will be contaminated. However given that the population is size 10,000 and the sample size is 10, the effect of one sample's success status on the next bottle's success status is negligible.
- ▶ Here the distribution of X is *approximately* Binomial:

$$X \sim \text{Binom}(10, 0.10)$$

where $\sim$ is read as “approximately distributed as”.

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If X has the binomial distribution with n observations and probability p of success on each observation, the possible values of X are $0, 1, 2, \dots, n$. If k is any one of these values,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- ▶ Read $\binom{n}{k}$ as “ n choose k ”. It counts the number of ways in which k successes can be arranged among n observations.
 - ▶ The binomial probability is this count multiplied by the probability of any one specific arrangement of the k successes.

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Definition: sampling distribution

A **sampling distribution** is the probability distribution of a statistic over all possible random samples of a fixed size. A simulation histogram can approximate it.

The distribution of the number of successes across many samples is called the **sampling distribution** for X .

Here, X is the number of successes in one sample.

The distribution of the proportion of successes across many samples is called the **sampling distribution** for $\hat{p} = X/n$.

Binomial approximation when N is much larger than n

Choose a simple random sample of size n from a population with proportion p of successes. When the population size (N) is much larger than the sample, the count X of successes in the sample has approximately the binomial distribution with parameters n and p .

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Example with two trials

In 1987, 29% of the adults in the United States smoked cigarettes, cigars, or pipes.

Let Y be a random variable that represents smoking status.

- ▶ $Y = 1$, an adult is currently a smoker
- ▶ $Y = 0$, an adult is not a current smoker

The two values of Smoking status are mutually exclusive and exhaustive.

What is the probability a randomly selected person is a smoker? $P(Y=1)$

What is the probability a randomly selected person is a non-smoker? $P(Y=0)$

Smoking status

Suppose that we randomly select two individuals from the population of adults in the United States.

The random variable X represents the number of persons in the pair who are current smokers.

First Person(Y_1)	Second Person (Y_2)	Probability	Number of Smokers (X)
0	0		0
1	0		1
0	1		1
1	1		2

Smoking status

Suppose that we randomly select two individuals from the population of adults in the United States.

The random variable X represents the number of persons in the pair who are current smokers.

First Person (Y_1)	Second Person (Y_2)	Probability	Number of Smokers (X)
0	0	$(1 - p) \times (1 - p)$	0
1	0	$p \times (1 - p)$	1
0	1	$(1 - p) \times p$	1
1	1	$p \times p$	2

Remember that we can multiply to get the probabilities here because the events are independent

Smoking status

Recall that for one trial, $P(\text{event}) + P(\text{no event}) = 1$

So $P(\text{smoker}) + P(\text{not smoker}) = 1$

We know $P(\text{smoker}) = 29\%$ so :

$1 - 0.29 = 0.71$ (probability of non smoker)

Calculate by hand using a table

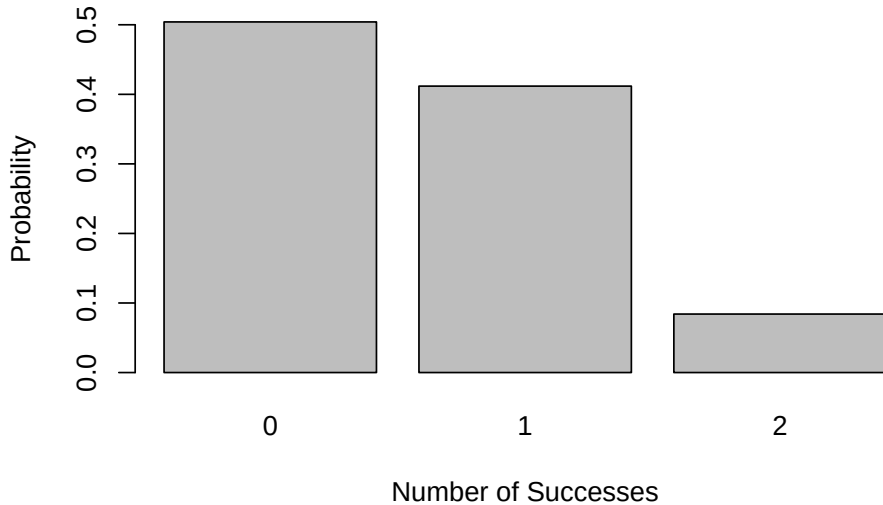
If 29% of US adults smoked, $p=0.29$, what are the values of these probabilities?

The random variable X represents the number of persons in the pair who are current smokers.

First Person (Y_1)	Second Person (Y_2)	Probability	Number of Smokers (X)
0	0	$(.71) \times (.71) = .5041$	0
1	0	$.29 \times (.71) = .2059$	1
0	1	$(.71) \times .29 = .2059$	1
1	1	$.29 \times .29 = .0841$	2

Probability distribution for 2 selected individuals

Probability Distribution



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In the binomial distribution the sum of all probabilities of potential outcomes equals 100% (1.0)

If you have a certain number of Bernoulli trials with probability of success (p),

What is the probability that X occurs at least once, $P(X \geq 1)$?

$$P(X \geq 1) = 1 - P(X = 0)$$

what is the probability that X occurs fewer than twice, $P(X < 2)$?

$$P(X < 2) = P(X = 1) + P(X = 0)$$

Binomial Probability Distributions: for n trials

What if we are interested in the expected outcomes if select a larger group of individuals? It starts to get cumbersome to write out that table by hand. The general expression of the probability distribution of a binomial random variable X where x is the number of successes in a sample of size n .

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

where $n = 1, 2, 3, \dots$ and $x = 0, 1, \dots, n$.

Binomial Combinations: How many combinations of n people give x successes?

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$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$\binom{2}{0} = \frac{2!}{0!(2-0)!} = 1$$

$$\binom{2}{1} = \frac{2!}{1!(2-1)!} = 2$$

**remember that $0! = 1$

So for 1 success in 2 individuals ($n=2$, $x=1$) where 29% are smokers ($p=0.29$)

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$P(X = 1) = \binom{2}{1} 0.29^1 (1 - 0.29)^{2-1} = 0.4118$$

We can use the formulas as shown to calculate the probability of a given number of successes from a binomial by hand

Or we could use R with `dbinom(x, size, prob)`.

This function calculates the probability of observing x successes when $X \sim \text{Binom}(n, p)$

```
dbinom(1,size=2,prob=0.29)
```

```
## [1] 0.4118
```

let's look further at sampling distributions using our container example...

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Sampling distribution of a count in R

First, set up a large population of size 10,000 where 10% of the containers are contaminated by benzene. We call benzene a “success” since it is coded as 1. We can see that 10% of the containers are contaminated and 1000 bottles are “successes”

We simulate these data:

```
container.id <- 1:10000  
benzene <- c(rep(0, 9000), rep(1, 1000))  
pop_data <- data.frame(container.id, benzene)
```

Sampling distribution of a count in R

```
# Population count and proportion contaminated by benzene  
pop_stats <- pop_data %>% summarize(pop_num_successes = sum(benzene),  
                                     pop_mean = mean(benzene))
```

```
pop_stats
```

```
##      pop_num_successes pop_mean  
## 1                1000      0.1
```

Sampling distribution of a count in R

Take a sample of size 10 from the population. Note that 10 is much smaller than 10,000.

- ▶ How many contaminated bottles are we expecting in the sample?
- ▶ Given that we sample 10, what is the full range of possible values we could see for X , the number of successes and p the proportion of successes?
- ▶ Which values are most likely?

```
# first sample
set.seed(1)
sample_data <- pop_data %>% sample_n(10)
sample_data %>% summarize(sample_num_successes = sum(benzene),
                          sample_mean = mean(benzene))
```

```
##   sample_num_successes sample_mean
## 1                      1         0.1
```

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Sampling distribution of a count in R

We only took one sample, and got 1 success for a sample mean of 10%. Is that usual or unusual?

To see what is most likely, we need to imagine repeatedly taking samples of size 10 from the population and calculating the sample number of successes and proportion of successes for each sample.

For the next few slides, we focus on the sampling distribution for X .

Sampling distribution of a count in R

The embedded code takes 1000 samples each of size 10.

It then calculates the mean sample proportion and number of successes for each sample and stores all the results in a data frame.

You don't need to know how the code works.

Sampling distribution of a count in R

Here are the first rows of the data frame we made on the previous slide. Each row represents an independent sample from the population.

```
head(many.sample.stats)
```

##	sample_proportion	sample_num_successes	sample.id
## 1	0.1	1	1
## 2	0.1	1	2
## 3	0.1	1	3
## 4	0.3	3	4
## 5	0.2	2	5
## 6	0.1	1	6

Sampling distribution of a count in R

We want to know: Of the 1000 samples, what proportion observed 0 contaminated bottles? What proportion observed 1 contaminated bottle? And so on. We can use `dplyr` functions to calculate this and plot the results in a histogram.

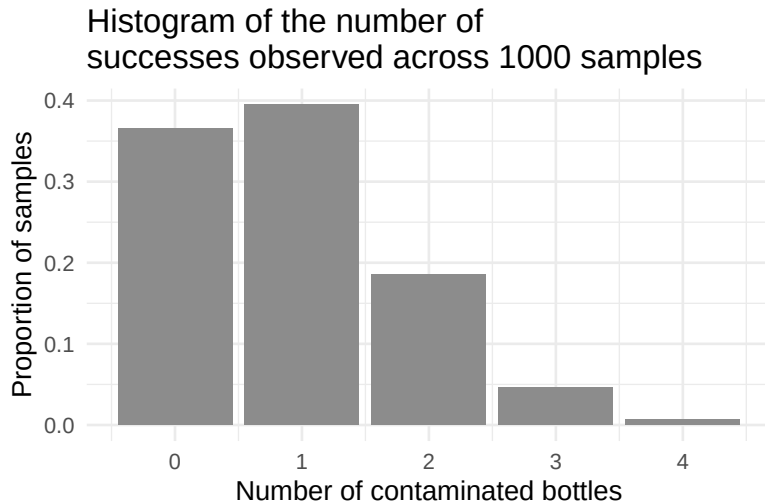
```
aggregated.stats <- many.sample.stats %>%  
  group_by(sample_num_successes) %>%  
  summarize(proportion = n()/1000, .groups = "drop")
```

Sampling distribution of a count in R

```
aggregated.stats
```

```
## # A tibble: 5 x 2
##   sample_num_successes proportion
##               <dbl>         <dbl>
## 1                   0         0.366
## 2                   1         0.395
## 3                   2         0.186
## 4                   3         0.046
## 5                   4         0.007
```

Sampling distribution of a count in R



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Sampling distribution of a count in R

As we will see in a moment, this histogram *approximates* the shape of the binomial distribution with $n = 10$ and $p = 0.1$. Observing one success is the most likely outcome. Why is that?

Worked probabilities, $x = 0$

We sampled $n=10$ bottles where the probability of success on any one pick is 10%.

- ▶ What is the chance of observing zero contaminated bottles?
- ▶ This means the first bottle is not contaminated and the second bottle is not contaminated, and ... and the tenth bottle is not contaminated

$$P(X_1 = 0 \text{ and } X_2 = 0 \text{ and } \dots \text{ and } X_{10} = 0)$$

$= P(X_1 = 0) \times P(X_2 = 0) \times \dots \times P(X_{10} = 0)$, using the multiplication rule for independent events

$$= (0.90)^{10}$$

$$= 0.3486784 = 34.9\%$$

Worked probabilities, $x = 1$

- ▶ What is the chance of observing exactly one contaminated bottle?
- ▶ Suppose that the first bottle was contaminated, then all the rest had to be not contaminated. What is the probability of observing this specific sequence of events?

$$\begin{aligned} &P(X_1 = 1 \text{ and } X_2 = 0 \text{ and } X_3 = 0 \text{ and } \dots \text{ and } X_{10} = 0) \\ &= P(X_1 = 1) \times P(X_2 = 0) \times P(X_3 = 0) \dots \times P(X_{10} = 0) \\ &= (0.1)^1 (0.90)^9 \\ &= 0.03874205 = 3.87\% \end{aligned}$$

But we're not done. This is only one specific way of observing exactly one contaminated bottle. What is another way? How many ways are there to observe exactly one contaminated bottle when there are ten bottles?

Worked probabilities, $x = 1$

There are ten ways to observe exactly one contaminated bottle:

- ▶ 1, 0, 0, 0, 0, 0, 0, 0, 0, 0
- ▶ 0, 1, 0, 0, 0, 0, 0, 0, 0, 0
- ▶ 0, 0, 1, 0, 0, 0, 0, 0, 0, 0
- ▶ 0, 0, 0, 1, 0, 0, 0, 0, 0, 0
- ▶ 0, 0, 0, 0, 1, 0, 0, 0, 0, 0
- ▶ ...
- ▶ 0, 0, 0, 0, 0, 0, 0, 0, 0, 1

Worked probabilities, $x = 1$

Remember

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$\binom{10}{1} = \frac{10!}{1!(10-1)!} = 10$$

Worked probabilities, $x = 1$

Each of these ten ways has the same probability of occurring.

Let A_j be the event that bottle j is the only contaminated bottle. The events $A_1, \dots, A_{10}$ are disjoint, so

$$P(X = 1) = \sum_{j=1}^{10} P(A_j) = 10(0.1)^1(0.9)^9.$$

$$= 10 \times (0.1)^1(0.9)^9$$

$$= 0.3874205 = 38.7\%$$

Worked probabilities, $x = 1$

We can check our calculations using the `dbinom()` function in R.

```
dbinom(x = 1, size = 10, prob = 0.1)
```

```
## [1] 0.3874205
```

This is exactly the answer we obtained.

Worked probabilities, $x = 2$

What is chance of observing exactly two contaminated bottles?

Following the same line of thinking, suppose that the first two bottles were contaminated. The chance of this happening is:

$$(0.1)^2(0.9)^8 = 0.004303672$$

But how many ways are there to observe exactly two contaminated bottles?

Worked probabilities, $x = 2$

Remember

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

$$\binom{10}{2} = \frac{10!}{2!(10-2)!} = 45$$

You could write out all the possibilities like last time, but there are a lot more!

Worked probabilities, $x = 2$

Note: we can get our calculators or R to perform this calculation for us. On our calculator, we need the button $\binom{n}{k}$, pronounced “n choose k”, and asks how many ways are there to have k successes when there are n individuals? In R we need the function `choose(n, k)`

```
choose(10, 2)
```

```
## [1] 45
```

There are 45 ways to observe exactly two contaminated bottles when you have ten bottles observed.

Make sure you can also perform this calculation on your calculator!

Worked probabilities, $x = 2$

To get the probability of observing exactly 2 contaminated bottles, because all of the possible combinations are equally possible, we can multiply 45 by the probability of observing the first two bottles as being contaminated:

$$45 \times (0.1)^2(0.9)^8 = 0.1937102 = 19.4\%$$

Check using R:

```
dbinom(x = 2, size = 10, prob = 0.1)
```

```
## [1] 0.1937102
```

All of the combinations with 10 bottles

Each of these is written as $\binom{10}{k}$, where k is 0, 1, 2, ..., 10. This is known as the **binomial coefficient**.

For $n = 10$, the coefficients are:

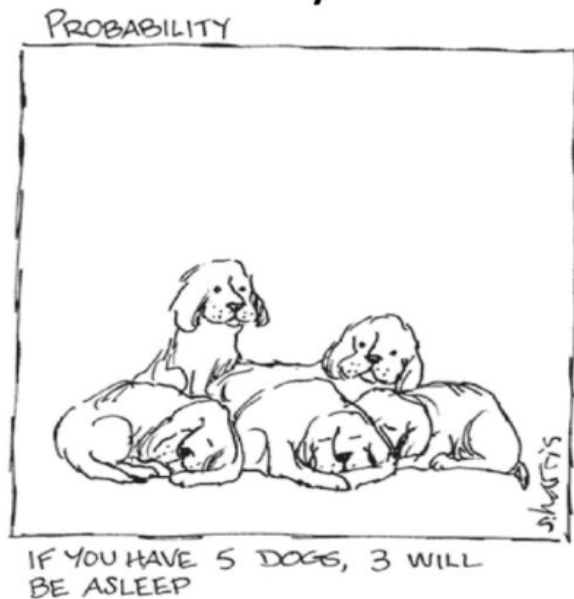
k	0	1	2	3	4	5	6	7	8	9	10
$\binom{10}{k}$	1	10	45	120	210	252	210	120	45	10	1

Notice the symmetric structure of $\text{choose}(n, k)$. Why is it symmetric?

New R code:

- ▶ `choose(n,k)` to calculate the number of combinations possible
- ▶ `dbinom(x,size,prob)` to calculate an exact discrete probability under a Binomial distribution

Comic Relief



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